

*“In theory, there is no difference
between
theory and practice. In practice,
there is.”*

— Yogi Berra (attributed)

Abstract

Abstract I

Multiagent AI systems—autonomous coding assistants, research pipelines, financial decision engines—have moved from prototype to production in under two years. With them comes a new class of security concern: attacks that target not data or infrastructure but the *reasoning processes* of AI agents. Prompt injections that propagate through delegation chains, trust relationships that launder adversarial influence, and coordination mechanisms vulnerable to strategic manipulation all represent cognitive attack surfaces absent from traditional security models.

Abstract II

The **Cognitive Integrity Framework (CIF)** is developed in **Parts 1, 2, and 4** of this series: formal treatment, running code and experiments, and cross-domain application. **Part 1** establishes mathematical foundations—a trust calculus with provably bounded delegation, defense composition algebras with multiplicative detection guarantees, and information-theoretic limits on attack stealth. **Part 2** provides computational validation: eight implemented defense modules (1,594 passing tests), a 950-attack corpus spanning four threat categories, and parametric architecture-aware simulation across four production multiagent topologies. **Part 4** applies the framework via the integrated CIF-AD-OODA model across ten critical operational domains.

Abstract III

This paper (Part 3) is a qualitative review and operator guide. It synthesizes those installments into accessible language, situates the formal results against current deployment practice, states what the research supports and what remains open, and gives practical recommendations for teams that build and run multiagent AI systems. No formal prerequisites are assumed; for proofs and definitions see Part 1, for empirical results Part 2, and for sector-specific use Part 4.

Paper Series I

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This is Part 3 of the four-part *Cognitive Security for Multiagent Operators* series:

- ▶ **Part 1** (DOI: 10.5281/zenodo.18364119): Formal foundations and theoretical analysis
- ▶ **Part 2** (DOI: 10.5281/zenodo.18364128): Computational validation and implementation
- ▶ **Part 3** (this paper): Qualitative review and practitioner's synthesis
- ▶ **Part 4:** Applications — CIF-AD-OODA integration and goal-hijacking defense across ten critical domains (rare-earth mining, nation-state alliances, cyber-security, drone warfare, supply chains, biowarfare, food security, trade wars, infrastructure, information ecosystems)